



**Eastern Mediterranean University
Computer Engineering Department**

**CMPE318/CMSE318
Principles of Programming Languages**

Assignment #1

Group #2

Team #9

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The Student Class:

```
Assignment1.py > ...
1  # CMPE 318 Assignment 1 - Updated Version
2
3  class Student:
4      def __init__(self, student_number, first_name, last_name, age, sex, major, nationality):
5          self.__student_number = student_number
6          self.__first_name = first_name
7          self.__last_name = last_name
8          self.__age = age
9          self.__sex = sex
10         self.__major = major
11         self.__nationality = nationality
12     ..
13     # Getters
14     def get_student_number(self):
15         | return self.__student_number
16
17     def get_first_name(self):
18         | return self.__first_name
19
20     def get_last_name(self):
21         | return self.__last_name
22
23     def get_age(self):
24         | return self.__age
25
26     def get_sex(self):
27         | return self.__sex
28
29     def get_major(self):
30         | return self.__major
31
32     def get_nationality(self):
33         | return self.__nationality
34
35     # Setters
36     def set_student_number(self, value):
37         | self.__student_number = value
38
39     def set_first_name(self, value):
40         | self.__first_name = value
41
42     def set_last_name(self, value):
43         | self.__last_name = value
44
45     def set_age(self, value):
46         | self.__age = value
47
48     def set_sex(self, value):
49         | self.__sex = value
50
51     def set_major(self, value):
52         | self.__major = value
53
54     def set_nationality(self, value):
55         | self.__nationality = value
56
57     def __str__(self):
58         | return (f"ID: {self.__student_number}, "
59         |         f"Name: {self.__first_name} {self.__last_name}, "
60         |         f"Age: {self.__age}, Sex: {self.__sex}, "
61         |         f"Major: {self.__major}, Nationality: {self.__nationality}")
62     ..
```

The Initialized List with 10 Students:

```
65 MAX_STUDENTS = 100
66 students = [Student(71165946,"Daniel","Mutua",22,"M","Computer Engineering","Kenya"),
67              Student(31908446,"Binwell","Kapila",21,"M","Artificial Intelligence Engineering","Zambia"),
68              Student(88228872,"Nour","Azza",19,"F","Software Engineering","Syria"),
69              Student(59094225,"Halima","Tanko",23,"F","Computer Intelligence Engineering","Nigeria"),
70              Student(45719442,"Syergyei","Khakimov",25,"M","Software Engineering","Uzbekistan"),
71              Student(87153631,"Kadir","Sert",25,"M","Artificial Engineering","Turkey"),
72              Student(27349117,"Chol","Puk",23,"M","Software Engineering","South Sudan"),
73              Student(36984691,"Meryem","Nacar",27,"F","Artificial Engineering","Cyprus"),
74              Student(16773424,"Mamo","Ayalew",28,"M","Computer Engineering","Ethiopia"),
75              Student(27471753,"Nadia","Babiker",21,"F","Software Engineering","Sudan")]
76
```

The Required Functions:

```
78 # Find student
79 def find_student(first_name, last_name):
80     for s in students:
81         if (s.get_first_name().lower() == first_name.lower() and
82             s.get_last_name().lower() == last_name.lower()):
83             return s
84     return None
85
86 # Add student
87 def add_student():
88     if len(students) >= MAX_STUDENTS:
89         print("Array full.")
90         return
91
92     student_number = input("Enter student number: ")
93     first_name = input("Enter first name: ")
94     last_name = input("Enter last name: ")
95
96     if find_student(first_name, last_name):
97         print("Student already exists.")
98         return
99
100     try:
101         age = int(input("Enter age: "))
102         if age < 0:
103             raise ValueError
104     except:
105         print("Invalid age.")
106         return
107
108     sex = input("Enter sex: ")
109     major = input("Enter major: ")
110     nationality = input("Enter nationality: ")
111
112     students.append(Student(student_number, first_name, last_name, age, sex, major, nationality))
113     print("Student added.")
114
```

```

117 # Show one student
118 def show_student():
119     first = input("First name: ")
120     last = input("Last name: ")
121
122     s = find_student(first, last)
123     if s:
124         print(s)
125     else:
126         print("Not found.")
127
128
129 # Show all
130 def show_all():
131     if not students:
132         print("No students.")
133         return
134
135     for s in students:
136         print(s)
137
138
139 # Age range
140 def show_age_range():
141     try:
142         min_age = int(input("Min age: "))
143         max_age = int(input("Max age: "))
144         if min_age < 0 or max_age < 0:
145             raise ValueError
146     except:
147         print("Invalid input.")
148         return
149
150     found = False
151     for s in students:
152         if min_age <= s.get_age() <= max_age:
153             print(s)
154             found = True
155
156     if not found:
157         print("No students in range.")
158
159
160 # Modify
161 def modify_student():
162     first = input("First name: ")
163     last = input("Last name: ")
164
165     s = find_student(first, last)
166     if not s:
167         print("Not found.")
168         return
169
170     print("1.ID 2.First 3.Last 4.Age 5.Sex 6.Major 7.Nationality")
171     choice = input("Choice: ")
172
173     if choice == "1":
174         s.set_student_number(input("New ID: "))
175     elif choice == "2":
176         s.set_first_name(input("New first name: "))
177     elif choice == "3":
178         s.set_last_name(input("New last name: "))
179     elif choice == "4":
180         try:
181             age = int(input("New age: "))
182             if age < 0:
183                 raise ValueError
184             s.set_age(age)
185         except:
186             print("Invalid.")
187     elif choice == "5":
188         s.set_sex(input("New sex: "))
189     elif choice == "6":
190         s.set_major(input("New major: "))
191     elif choice == "7":
192         s.set_nationality(input("New nationality: "))
193     else:
194         print("Invalid choice.")
195

```

```

197 # Delete
198 def delete_student():
199     first = input("First name: ")
200     last = input("Last name: ")
201
202     s = find_student(first, last)
203     if s:
204         students.remove(s)
205         print("Deleted.")
206     else:
207         print("Not found.")
208
209
210 # Write file
211 def write_file():
212     filename = input("Filename: ")
213
214     with open(filename, "w") as f:
215         for s in students:
216             f.write(f"{s.get_student_number()}, {s.get_first_name()}, {s.get_last_name()}, "
217                 f"{s.get_age()}, {s.get_sex()}, {s.get_major()}, {s.get_nationality()}\n")
218
219     print("Saved.")
220
221
222 # Read file
223 def read_file():
224     filename = input("Filename: ")
225
226     try:
227         with open(filename, "r") as f:
228             students.clear()
229             for line in f:
230                 data = line.strip().split(",")
231                 if len(data) == 7:
232                     students.append(Student(data[0], data[1], data[2],
233                                             int(data[3]), data[4], data[5], data[6]))
234             print("Loaded.")
235     except:
236         print("Error reading file.")
237
238
239 # Bonus search
240 def search_any():
241     keyword = input("Enter keyword: ").lower()
242
243     found = False
244     for s in students:
245         if (keyword in s.get_first_name().lower() or
246             keyword in s.get_last_name().lower() or
247             keyword in s.get_major().lower() or
248             keyword in s.get_nationality().lower() or
249             keyword==s.get_age().__str__() or
250             keyword==s.get_student_number().__str__()):
251             print(s)
252             found = True
253
254     if not found:
255         print("No matches.")
256

```

The Menu:

```
258 def menu():
259     while True:
260         print("\n===== STUDENT MANAGEMENT SYSTEM =====")
261         print("1. Add a new student")
262         print("2. Find a student by first name and last name")
263         print("3. Show all students")
264         print("4. Show all students within an age range")
265         print("5. Modify a student record")
266         print("6. Delete a student")
267         print("7. Write student data to a file")
268         print("8. Read student data from a file")
269         print("9. Find students by any criterion (Bonus)")
270         print("10. Quit")
271         print("=====")
272
273         choice = input("Enter your choice (1-10): ")
274
275         if choice == "1":
276             add_student()
277         elif choice == "2":
278             show_student()
279         elif choice == "3":
280             show_all()
281         elif choice == "4":
282             show_age_range()
283         elif choice == "5":
284             modify_student()
285         elif choice == "6":
286             delete_student()
287         elif choice == "7":
288             write_file()
289         elif choice == "8":
290             read_file()
291         elif choice == "9":
292             search_any()
293         elif choice == "10":
294             print("Exiting program... Goodbye!")
295             break
296         else:
297             print("Invalid choice. Please enter a number between 1 and 10.")
298
299
300 menu()
```


Program Output:

```
===== STUDENT MANAGEMENT SYSTEM =====
1. Add a new student
2. Find a student by first name and last name
3. Show all students
4. Show all students within an age range
5. Modify a student record
6. Delete a student
7. Write student data to a file
8. Read student data from a file
9. Find students by any criterion (Bonus)
10. Quit
=====
Enter your choice (1-10): █
```

The outputted menu with all the choices available.

Option 3: Showing all the students

```
Enter your choice (1-10): 3
ID: 71165946, Name: Daniel Mutua, Age: 22, Sex: M, Major: Computer Engineering, Nationality: Kenya
ID: 31908446, Name: Binwell Kapila, Age: 21, Sex: M, Major: Artificial Intelligence Engineering, Nationality: Zambia
ID: 88228872, Name: Nour Azza, Age: 19, Sex: F, Major: Software Engineering, Nationality: Syria
ID: 59094225, Name: Halima Tanko, Age: 23, Sex: F, Major: Computer Intelligence Engineering, Nationality: Nigeria
ID: 45719442, Name: Syergyeyi Khakimov, Age: 25, Sex: M, Major: Software Engineering, Nationality: Uzbekistan
ID: 87153631, Name: Kadir Sert, Age: 25, Sex: M, Major: Artificial Engineering, Nationality: Turkey
ID: 27349117, Name: Chol Puk, Age: 23, Sex: M, Major: Software Engineering, Nationality: South Sudan
ID: 36984691, Name: Meryem Nacar, Age: 27, Sex: F, Major: Artificial Engineering, Nationality: Cyprus
ID: 16773424, Name: Mamo Ayalew, Age: 28, Sex: M, Major: Computer Engineering, Nationality: Ethiopia
ID: 27471753, Name: Nadia Babiker, Age: 21, Sex: F, Major: Software Engineering, Nationality: Sudan
```

The program shows the initial list of 10 students along with their details.

Option 1: Adding a student

```
Enter your choice (1-10): 1
Enter student number: 12345678
Enter first name: John
Enter last name: Doe
Enter age: 25
Enter sex: M
Enter major: Computer Engineering
Enter nationality: United States
Student added.
```

The program asks you to enter the student's details one by one, and then shows a confirmation message that the student has been added.

```
Enter your choice (1-10): 3
ID: 71165946, Name: Daniel Mutua, Age: 22, Sex: M, Major: Computer Engineering, Nationality: Kenya
ID: 31908446, Name: Binwell Kapila, Age: 21, Sex: M, Major: Artificial Intelligence Engineering, Nationality: Zambia
ID: 88228872, Name: Nour Azza, Age: 19, Sex: F, Major: Software Engineering, Nationality: Syria
ID: 59094225, Name: Halima Tanko, Age: 23, Sex: F, Major: Computer Intelligence Engineering, Nationality: Nigeria
ID: 45719442, Name: Syergyeyi Khakimov, Age: 25, Sex: M, Major: Software Engineering, Nationality: Uzbekistan
ID: 87153631, Name: Kadir Sert, Age: 25, Sex: M, Major: Artificial Engineering, Nationality: Turkey
ID: 27349117, Name: Chol Puk, Age: 23, Sex: M, Major: Software Engineering, Nationality: South Sudan
ID: 36984691, Name: Meryem Nacar, Age: 27, Sex: F, Major: Artificial Engineering, Nationality: Cyprus
ID: 16773424, Name: Mamo Ayalew, Age: 28, Sex: M, Major: Computer Engineering, Nationality: Ethiopia
ID: 27471753, Name: Nadia Babiker, Age: 21, Sex: F, Major: Software Engineering, Nationality: Sudan
ID: 12345678, Name: John Doe, Age: 25, Sex: M, Major: Computer Engineering, Nationality: United States
```

The updated list including the last added student.

Option 2: Finding a student by name

```
Enter your choice (1-10): 2
First name: Nour
Last name: Azza
ID: 88228872, Name: Nour Azza, Age: 19, Sex: F, Major: Software Engineering, Nationality: Syria
```

When given a student's first and last names, the program returns the full details of the student.

Option 4: Showing all students in a certain age range

```
Enter your choice (1-10): 4
Min age: 20
Max age: 23
ID: 71165946, Name: Daniel Mutua, Age: 22, Sex: M, Major: Computer Engineering, Nationality: Kenya
ID: 31908446, Name: Binwell Kapila, Age: 21, Sex: M, Major: Artificial Intelligence Engineering, Nationality: Zambia
ID: 59094225, Name: Halima Tanko, Age: 23, Sex: F, Major: Computer Intelligence Engineering, Nationality: Nigeria
ID: 27349117, Name: Chol Puk, Age: 23, Sex: M, Major: Software Engineering, Nationality: South Sudan
ID: 27471753, Name: Nadia Babiker, Age: 21, Sex: F, Major: Software Engineering, Nationality: Sudan
```

When given a minimum age and a maximum age, the program returns only students whose ages fall within that range.

Option 5: Modifying a student's details

```
Enter your choice (1-10): 5
First name: Syergyeyi
Last name: Khakimov
1.ID 2.First 3.Last 4.Age 5.Sex 6.Major 7.Nationality
Choice: 6
New major: Artificial Intelligence Engineering
```

The program asks for the first and last names of the student whose details will be modified, and then asks for which detail to modify.

```
Enter your choice (1-10): 2
First name: Syergyeyi
Last name: Khakimov
ID: 45719442, Name: Syergyeyi Khakimov, Age: 25, Sex: M, Major: Artificial Intelligence Engineering, Nationality: Uzbekistan
```

The student's major has been updated to the new value.

Option 6: Deleting a student

```
Enter your choice (1-10): 6
First name: Nadia
Last name: Babiker
Deleted.
```

The program asks for the first and last names of the student to be deleted, then deletes them.

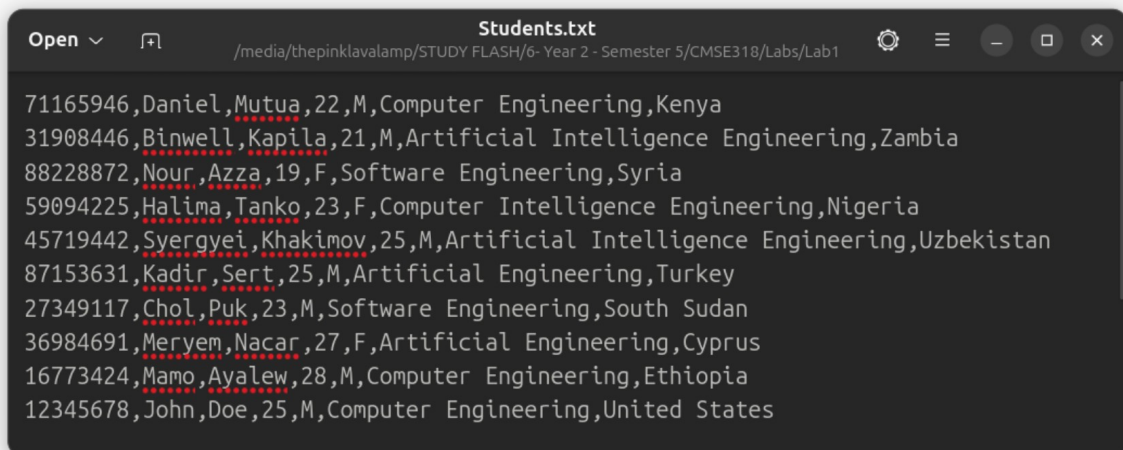
```
Enter your choice (1-10): 3
ID: 71165946, Name: Daniel Mutua, Age: 22, Sex: M, Major: Computer Engineering, Nationality: Kenya
ID: 31908446, Name: Binwell Kapila, Age: 21, Sex: M, Major: Artificial Intelligence Engineering, Nationality: Zambia
ID: 88228872, Name: Nour Azza, Age: 19, Sex: F, Major: Software Engineering, Nationality: Syria
ID: 59094225, Name: Halima Tanko, Age: 23, Sex: F, Major: Computer Intelligence Engineering, Nationality: Nigeria
ID: 45719442, Name: Syergyeyi Khakimov, Age: 25, Sex: M, Major: Artificial Intelligence Engineering, Nationality: Uzbekistan
ID: 87153631, Name: Kadir Sert, Age: 25, Sex: M, Major: Artificial Engineering, Nationality: Turkey
ID: 27349117, Name: Chol Puk, Age: 23, Sex: M, Major: Software Engineering, Nationality: South Sudan
ID: 36984691, Name: Meryem Nacar, Age: 27, Sex: F, Major: Artificial Engineering, Nationality: Cyprus
ID: 16773424, Name: Mamo Ayalew, Age: 28, Sex: M, Major: Computer Engineering, Nationality: Ethiopia
ID: 12345678, Name: John Doe, Age: 25, Sex: M, Major: Computer Engineering, Nationality: United States
```

The updated list with the deleted student missing.

Option 7: Writing student data to a file


```
Enter your choice (1-10): 7
Filename: Students.txt
Saved.
```

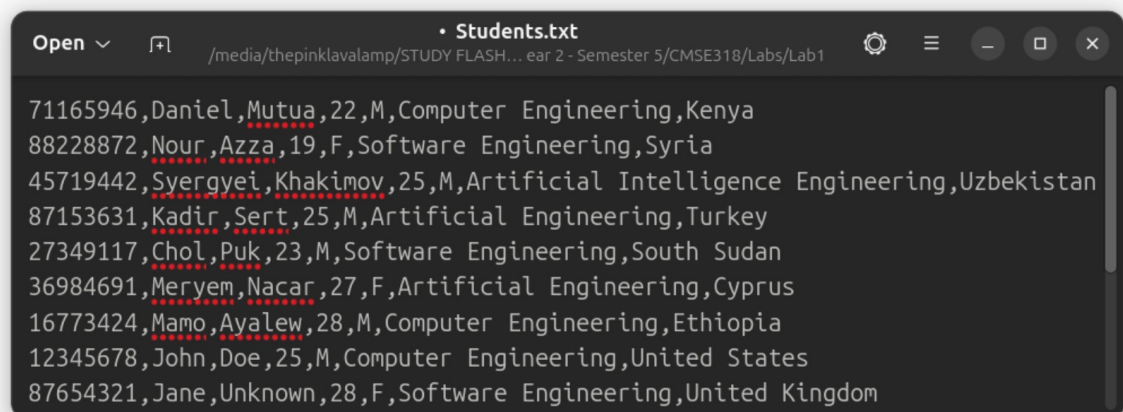
The program asks for a file name. It then creates a file with that name and saves the student data inside it.



```
Open ▾ /media/thespinklavalamp/STUDY FLASH/6- Year 2 - Semester 5/CMSE318/Labs/Lab1
71165946,Daniel,Mutua,22,M,Computer Engineering,Kenya
31908446,Binwell,Kapila,21,M,Artificial Intelligence Engineering,Zambia
88228872,Nour,Azza,19,F,Software Engineering,Syria
59094225,Halima,Tanko,23,F,Computer Intelligence Engineering,Nigeria
45719442,Syergyei,Khakimov,25,M,Artificial Intelligence Engineering,Uzbekistan
87153631,Kadir,Sert,25,M,Artificial Engineering,Turkey
27349117,Chol,Puk,23,M,Software Engineering,South Sudan
36984691,Meryem,Nacar,27,F,Artificial Engineering,Cyprus
16773424,Mamo,Ayalew,28,M,Computer Engineering,Ethiopia
12345678,John,Doe,25,M,Computer Engineering,United States
```

The file created with the student data saved inside it.

Option 8: Reading student data from a file



```
Open ▾ /media/thespinklavalamp/STUDY FLASH... ear 2 - Semester 5/CMSE318/Labs/Lab1
71165946,Daniel,Mutua,22,M,Computer Engineering,Kenya
88228872,Nour,Azza,19,F,Software Engineering,Syria
45719442,Syergyei,Khakimov,25,M,Artificial Intelligence Engineering,Uzbekistan
87153631,Kadir,Sert,25,M,Artificial Engineering,Turkey
27349117,Chol,Puk,23,M,Software Engineering,South Sudan
36984691,Meryem,Nacar,27,F,Artificial Engineering,Cyprus
16773424,Mamo,Ayalew,28,M,Computer Engineering,Ethiopia
12345678,John,Doe,25,M,Computer Engineering,United States
87654321,Jane,Unknown,28,F,Software Engineering,United Kingdom
```

The list in the file was manually modified slightly to show the difference in the program.

```
Enter your choice (1-10): 8
Filename: Students.txt
Loaded.
```

The program asks for the name of the file to be read and then reads the data from it.

```
Enter your choice (1-10): 3
ID: 71165946, Name: Daniel Mutua, Age: 22, Sex: M, Major: Computer Engineering, Nationality: Kenya
ID: 88228872, Name: Nour Azza, Age: 19, Sex: F, Major: Software Engineering, Nationality: Syria
ID: 45719442, Name: Syergyey Khakimov, Age: 25, Sex: M, Major: Artificial Intelligence Engineering, Nationality: Uzbekistan
ID: 87153631, Name: Kadir Sert, Age: 25, Sex: M, Major: Artificial Engineering, Nationality: Turkey
ID: 27349117, Name: Chol Puk, Age: 23, Sex: M, Major: Software Engineering, Nationality: South Sudan
ID: 36984691, Name: Meryem Nacar, Age: 27, Sex: F, Major: Artificial Engineering, Nationality: Cyprus
ID: 16773424, Name: Mamo Ayalew, Age: 28, Sex: M, Major: Computer Engineering, Nationality: Ethiopia
ID: 12345678, Name: John Doe, Age: 25, Sex: M, Major: Computer Engineering, Nationality: United States
ID: 87654321, Name: Jane Unknown, Age: 28, Sex: F, Major: Software Engineering, Nationality: United Kingdom
```

The updated list loaded from the file.

Option 9: Finding students by any criterion

```
Enter your choice (1-10): 9
Enter keyword: Ethiopia
ID: 16773424, Name: Mamo Ayalew, Age: 28, Sex: M, Major: Computer Engineering, Nationality: Ethiopia
```

The program asks for a keyword (such as a specific age, name, nationality, etc...) and checks if any of the fields contain any match.

```
Enter your choice (1-10): 9
Enter keyword: Software Engineering
ID: 88228872, Name: Nour Azza, Age: 19, Sex: F, Major: Software Engineering, Nationality: Syria
ID: 27349117, Name: Chol Puk, Age: 23, Sex: M, Major: Software Engineering, Nationality: South Sudan
ID: 87654321, Name: Jane Unknown, Age: 28, Sex: F, Major: Software Engineering, Nationality: United Kingdom

Enter your choice (1-10): 9
Enter keyword: 25
ID: 45719442, Name: Syergyey Khakimov, Age: 25, Sex: M, Major: Artificial Intelligence Engineering, Nationality: Uzbekistan
ID: 87153631, Name: Kadir Sert, Age: 25, Sex: M, Major: Artificial Engineering, Nationality: Turkey
ID: 12345678, Name: John Doe, Age: 25, Sex: M, Major: Computer Engineering, Nationality: United States
```

Option 10: Exiting the program

```
Enter your choice (1-10): 10
Exiting program... Goodbye!
```